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Homework 2: Markov Decision Processes Smart Home Energy Management

1. Bellman Expectation

In an MDP where the goal is to minimize cost, the action-value function (or Cost-to-Go) $Q(s,a)$ is given by the Bellman Expectation equation for costs:

$$Q(s,a) = C(s,a) + \gamma \sum_{s'} P(s'|s,a)V(s')$$

Where:

- $C(s,a)$ is the immediate cost of taking action a in state s.
- γ is the discount factor (given as 0.9).
- $P(s'|s,a)$ is the probability of transitioning to state s' given state s and action a.
- $V(s')$ is the estimated value (future cost) of the next state s'.

For $Q(S_1, a_B)$ we must calculate the expected cost-to-go when choosing to use the battery while in the charged state. Using the provided transition matrix P_{aB}, the probabilities of transitioning from S_1 are $P(S_0|S_1, a_B) = 0.7$ and $P(S_1|S_1, a_B) = 0.3$.

Expanding the Bellman equation for this specific state and action:

$$Q(S_1, a_B) = C(S_1, a_B) + \gamma \left[P(S_0|S_1, a_B)V(S_0) + P(S_1|S_1, a_B)V(S_1)\right]$$

Substituting the known values ($C(S_1, a_B) = 0$, $\gamma = 0.9$, $V(S_0) = 100$, and $V(S_1) = 20$):

$$Q(S_1, a_B) = 0 + 0.9 \times [(0.7 \times 100) + (0.3 \times 20)]$$

$$Q(S_1, a_B) = 0 + 0.9 \times [70 + 6]$$

$$Q(S_1, a_B) = 0 + 0.9 \times 76$$

$$Q(S_1, a_B) = 68.4$$

Now, let's plug the values into the Bellman equation for action a_G:

$$Q(S_1, a_G) = C(S_1, a_G) + \gamma \left[P(S_0|S_1, a_G)V(S_0) + P(S_1|S_1, a_G)V(S_1)\right]$$

Substitute the known values ($C(S_1, a_G) = 10$, $\gamma = 0.9$, $V(S_0) = 100$, $V(S_1) = 20$):

$$Q(S_1, a_G) = 10 + 0.9 \times [(0.1 \times 100) + (0.9 \times 20)]$$

$$Q(S_1, a_G) = 10 + 0.9 \times [10 + 18]$$

$$Q(S_1, a_G) = 10 + 0.9 \times [28]$$

$$Q(S_1, a_G) = 10 + 25.2$$

$$Q(S_1, a_G) = 35.2$$

Because the objective is to minimize the cost, the optimal policy for this iteration will choose the action that yields the lowest expected cost Q.

Comparing the two results:

- $Q(S_1, a_G) = 35.2$
- $Q(S_1, a_B) = 68.4$

Since $35.2 < 68.4$, the optimal action for state S_1 in this iteration is a_G (UseGrid).

2. Policy Evaluation

For a fixed policy π, the matrix form of the Bellman Expectation equation is:

$$V^\pi = C^\pi + \gamma P^\pi V^\pi$$

Given the policy to always use the Grid (a_G), we define our matrices and vectors as follows:

- The discount factor: $\gamma = 0.9$
- The cost vector for a_G: $C^\pi = \begin{bmatrix} 10 \\ 10 \end{bmatrix}$
- The transition matrix for a_G: $P^\pi = \begin{bmatrix} 0.3 & 0.7 \\ 0.1 & 0.9 \end{bmatrix}$
- The value vector we want to solve for: $V^\pi = \begin{bmatrix} V(S_0) \\ V(S_1) \end{bmatrix}$

Plugging these into the matrix equation:

$$\begin{bmatrix} V(S_0) \\ V(S_1) \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix} + 0.9 \begin{bmatrix} 0.3 & 0.7 \\ 0.1 & 0.9 \end{bmatrix} \begin{bmatrix} V(S_0) \\ V(S_1) \end{bmatrix}$$

After the matrix multiplication, we get two new linear equations:

$$V(S_0) = 10 + 0.9 \times [0.3V(S_0) + 0.7V(S_1)]$$

$$V(S_1) = 10 + 0.9 \times [0.1V(S_0) + 0.9V(S_1)]$$

Solving these two equations, we obtain the following result:

$$V(S_0) = 100$$

$$V(S_1) = 100$$

This makes perfect logical sense because the cost of using the grid is always 10 regardless of the state, the agent incurs a cost of 10 every single step into infinity. The sum of an infinite geometric series with an immediate cost of 10 and a discount factor of 0.9 is exactly $10/(1 - 0.9) = 100$.

Problem 3: Contraction Mappings

3.1

We start with the definition of the Bellman Optimality Operator:

$$(TV_{new})(s) = \min_a \left[C(s,a) + \gamma \sum_{s'} P(s'|s,a) V_{new}(s') \right]$$

We use the fact that $V_{new}(s') = V_{old}(s') + c$:

$$(TV_{new})(s) = \min_a \left[C(s,a) + \gamma \sum_{s'} P(s'|s,a)(V_{old}(s') + c) \right] \iff$$

$$\iff (TV_{new})(s) = \min_a \left[C(s,a) + \gamma \sum_{s'} P(s'|s,a) V_{old}(s') + \gamma \sum_{s'} P(s'|s,a) c \right]$$

Because $P(s'|s,a)$ represents a valid probability distribution of transitioning to the next states, the sum of all probabilities must equal exactly 1. Hence,

$$(TV_{new})(s) = \min_a \left[C(s,a) + \gamma \sum_{s'} P(s'|s,a) V_{old}(s') + \gamma c \right] \iff$$

$$\iff (TV_{new})(s) = \min_a \left[C(s,a) + \gamma \sum_{s'} P(s'|s,a) V_{old}(s') \right] + \gamma c \iff$$

$$\iff (TV_{new})(s) = (TV_{old})(s) + \gamma c$$

3.2

The property established in above is exactly what makes the Bellman operator a contraction mapping.

Considering the maximum possible difference between any two arbitrary value functions, U and V, across all states. Let this maximum difference be c. This means $U(s) \leq V(s) + c$.

Applying the Bellman operator T to both sides yields $TU \leq T(V + c)$. Using the property proved in 3.1, this expands to $TU \leq TV + \gamma c$. This means the new maximum distance between the value functions after one update is bounded by γc. Since $\gamma = 0.9$, which is strictly less than 1, the distance between any two value functions shrinks by at least 10% with every single iteration.

According to the Banach Fixed-Point Theorem, if an operator always shrinks the distance between points, repeatedly applying it will inevitably pull all starting points toward exactly one unique destination.

Therefore, no matter what initial V_0 is chosen, the Value Iteration algorithm will consistently and successfully converge to the single, optimal fixed point.